01 · POWER

PACK / PROTECTION / DISCONNECT

SW1 is now the emergency stop — mount where a hand lands

without looking. The e-stop circuit itself was removed.

BT1 12x INR18650 2200mAh, 4S3P. 14.8V nom, 12.0–16.8V window.

BT1
4S3P 97.7Wh

BMS1
6S 50A

F1
20A

SW1
MAIN

J1 taps the pack side of BMS1 so the balance

charger sees the cells directly.

J1
charge

FLAG1
PWR_FLAG

Re-run this trunk in AWG14 — AWG18 is 16% of the copper

it feeds across the 4 motor branches on sheet 2.

V_PACK 16.8 V nominal

Isolates the motors only — Pi + sensors stay live

5 V AND 6 V RAILS

MOTOR RAIL — SWITCHED SEPARATELY

FLAG2
PWR_FLAG

SW2
MOTOR

V_MOTOR

U1
5.0V

U2
6.0V

V_5V

V_SERVO

C2
2200u

CC pot at MAX, set 5.00 V

CC pot at MAX, set 6.00 V

cap at the servo plug too

→ 4x BTS7960 on sheet 2

Pi SUPPLY — ISOLATED FROM THE PACK

Not on the pack. Motor current spikes cannot sag a rail

the Pi is not connected to.

BANK1
10000mAh 20W

V_PI

USB to the ESP32 for power and serial — the only GND tie

back to the pack side.

02 · DRIVETRAIN

ESP32 — MOTOR + SERVO CONTROL ONLY

Powered from the Pi's USB port ONLY.

Do not also feed 5V into VIN — two supplies, one rail.

10k pulldown holds MOT_EN low until the ESP32

drives it — no glitch during the boot float.

4x BTS7960 → 4x GEARMOTOR

MOT_EN

V_3V

GND

L_RPWM

L_LPWM

R_RPWM

R_LPWM

R1
10R

Q1

Q2

Q3

Q4

M1
FL 12V gear

M2
FR 12V gear

M3
RR 12V gear

M4
RR 12V gear

M+/M− drop clear, jog across, enter the motor from directly above.

U3
ESP32-WROOM-32

USB_FROM_PI

GPIO25_L_RPWM
GPIO26_L_LPWM
GPIO27_R_RPWM
GPIO14_R_LPWM
GPIO23_MOT_EN
GPIO18_SERVO_PAN
GPIO21_SERVO_TILT
GPIO34_VPACK_DIV
GPIO39_ENC_L
GPIO32_ENC_R

not fitted
6-wire motors only
6-wire motors only

SERVOS

SV1
pan

SV2
tilt

V_SERVO

V+ from V_SERVO (6V rail) — NOT the 5V logic rail

03 · SENSORS

RASPBERRY PI 5

Powered from BANK1 on sheet 1, not the main pack.

I2C SENSOR BUS

R4
4.7k

R5
4.7k

2x 4.7k pull-ups, Pi end only. Power every breakout from

3V3, never 5V — cheap boards pull SDA/SCL up to their own VCC.

AMG8833 AD_SEL → 3V3 (0x69)

MPU6050 AD0 → GND (0x68)

GND

V_PI_3V3

U4
AMG8833
addr 0x69

U5
OAT31
addr 0x44

U6
MPU6050
addr 0x68

U7
INA219
addr 0x40

U8
MLX90614
addr 0x5A

Pi1
Raspberry Pi 5

V_PI

LED_IN1
LED_IN2
LED_IN3
LED_IN4

XSHUT_A
XSHUT_B
SHUTDOWN_BTN
USB_FROM_PI

SDA

SCL

Camera: CSI ribbon, not I2C. Pi 5 needs the narrow

22-pin adapter cable for Camera Module 3.

GPIO26 → momentary button → GND. Clean shutdown —

a hard power cut corrupts the SD card.

ToF RANGING — ON THE SENSOR ARM

U9
vl53l1x (arm)

U10
vl53l1x (arm)

XSHUT_A

XSHUT_B

both ship at 0x29 — reassigned to 0x30/0x31 by XSHUT,

EVERY boot. Mean = standoff, difference = yaw to the wall.

LED RING — ONE AT A TIME

fired ONE at a time — two lit at once breaks the

photometric solve. 220R sets LED current from V_5V.

R6
220R

R7
220R

R8
220R

R9
220R

D1
photometric

D2
photometric

D3
photometric

D4
photometric

U11
ULN2803A

LED_IN1
LED_IN2
LED_IN3
LED_IN4
V_5V

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